

Filippo Merlo

MSc in Cognitive Science and Computational Modeling of Language and Cognition



About me

I have obtained an MSc in Cognitive Science and Natural Language Processing, with a keen passion for machine learning and psychology. My expertise lies in model evaluation and experimental design, encompassing both human and artificial intelligence models. Through numerous research internships, I have gained substantial experience in conducting research projects and fostering teamwork. I am confident that my curiosity, researching skills and multidisciplinary background would make me a valuable addition to the team!

Areas of specialization

Natural Language Processing

- Vision Language Models
- Cognitive Psychology

References

Professor Albert Gatt

Professor in Natural Language Generation at Utrecht University, Utrecht
a.gatt@uu.nl

Professor Raffaella Bernardi

Professor in Computational Linguistic at Free University of Bolzano, Bozen
Raffaella.Bernardi@unibz.it

Dr. Ece Takmaz

Postdoctoral Researcher in Natural Language Processing, Utrecht University e.k.takmaz@uu.nl

 [filippo-merlo.github.io](https://github.com/filippo-merlo)

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WORKING EXPERIENCE

Mar 2024 – Mar 2025

Internship at Natural Language Processing group

RESEARCH · Utrecht-Netherlands



Master thesis research project on State of the art Vision-Language Models (VLMs) Evaluation. Dataset exploration and data curation, Model training and finetuning with PyTorch, VLMs (CLIP, BLIP-3, LLaVa-OneVision, Molmo), Vision Models (ViT, YOLO), Image Generative Models (Stable Diffusion, PowerPaint).

Sept 2023 – Feb 2024

Internship at Language, Interaction and Computation lab

RESEARCH · Trento-Italy



Research project on Evaluating Generative Models' Reasoning Abilities. Dataset exploration (WordNet, ConceptNet, smaller norm datasets) and data curation. Prompt Engineering and Multi-agents experimental setting.

Oct 2023 – Jan 2024

University Tutor in Computational Linguistics

TEACHING · Trento-Italy



Tutoring students in the Computational Linguistics course, teaching foundational concepts and programming laboratory (Python, Hugging Face) and overseeing the final projects.

Dec 2020- Feb 2022

Internship at Department of General Psychology

RESEARCH · Padova-Italy



(Sett 21-Feb 22) Thesis Project on Online Behavioural Studies on human motion perception. Online Data Collection with JsPsych and Amazon Mechanical Turk. Data analysis with R.

(Dec 20-Jun 21) Internship project on online Behavioural studies during the Covid-19 era. Programming in HTML/CSS/JavaScript.

SKILLS

Languages: Python, Javascript, HTML, CSS

Statistics: R, MATLAB

Technologies: PyTorch, Scikit-learn, Tensorflow, Hugging Face, OpenCV, PyGame, jsPsych, Psychtoolbox-3.

PUBLICATIONS

2025 Filippo Merlo, Ece Takmaz, Wenkai Chen, Albert Gatt, 2025. COOCO – Common Objects Out-of-Context – Semantic Violation in Scenes: Investigating Multimodal Context in Referential Communication. *arXiv:2506.22274*

2023 Leonardo Bertolazzi, Davide Mazzaccara, Filippo Merlo, and Raffaella Bernardi. 2023. ChatGPT's Information Seeking Strategy: Insights from the 20-Questions Game. In *Proceedings of the 16th International Natural Language Generation Conference*, pages 153–162, Prague, Czechia. Association for Computational Linguistics., Association for Computational Linguistics.

AWARDS

2023 Best Evaluation Paper at INLG 2023 (<https://sigdialinlg2023.github.io/awards.html>)

EDUCATION

Sept 2022 – Mar 2025

M.Sc. in Cognitive Science - Computational Modelling of Language and Cognition.

TRENTO · Italy 📍

GPA: 29.5/30, Final Evaluation: 110 cum laude/110



Sept 2021 – Aug 2022

M.Sc. in Neuroscience

PADOVA-WARSAWA · Italy-Poland 📍

GPA: 29.9/30, Final Evaluation: Discontinued



Sept 2018 – Sept 2021

B.Sc. in Cognitive and Psychobiological Science

PADOVA · Italy 📍

GPA: 28.9/30, Final Evaluation: 110 cum laude/110



WORKSHOPS

- **NLG in the Lowlands, 2025**, Tilburg University. Poster presentation. <https://sites.google.com/view/nlg-in-the-lowlands-2025/home>
- **CORE Project Workshop 2024**, Universitat Pompeu Fabra. Poster presentation. <https://www.upf.edu/web/glif/2024-core-workshop/program>
- **European Summer School on Artificial Intelligence (ESSAI) 2023**. Student.
- **Lectures on Computational Linguistics 2023**, University of Pisa. Poster presentation.

LANGUAGES

Italian	C2	mother tongue
English	C2	working proficiency

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